

AROMA

AI-powered Repository & Object Model Analyzer

Project Overview & Pitch

From idea to impact in 48 hours

IBM Bob Hackathon 2026 - Turn idea into impact faster
Built with IBM Bob IDE | Powered by IBM watsonx.ai Granite

The Problem

Developers spend 30-50% of their time understanding code rather than building. Legacy codebases are painful: no documentation, hidden dependencies, no test coverage, security blind spots, and constant context switching.

Our Solution: AROMA

AROMA (AI-powered Repository & Object Model Analyzer) is a comprehensive web-based development tool that helps developers understand, maintain, and improve legacy codebases using AI. It addresses ALL five pain points in a single, unified web application.

Hackathon Alignment

Challenge:	Turn idea into impact faster
Speed:	Get up to speed on existing code quickly
Docs:	Generate documentation and tests automatically
Effort:	Reduce effort spent on repetitive tasks

IBM Technology Stack

AI Inference:	IBM watsonx.ai Granite 4-h-small
Auth:	IBM IAM bearer token (auto-refresh)
Dev Partner:	IBM Bob IDE / Bob Shell (5 sessions)
Framework:	Next.js 15 + React 19 + TypeScript 5

Key Design Decisions

- Web application - accessible anywhere, no install
- 6 specialized modules - deep functionality per task
- Manual code input - privacy-preserving
- AI-first with fallbacks - always responsive
- IBM watsonx.ai backend - official IBM AI platform